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Technical Specification

O-RAN Work Group 2 (Non-RT RIC & A1/R1 interface)

Service Management and Exposure SMO Service: General Aspects, Principles, Use Cases and Requirements

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Contents

Foreword....4
Modal verbs terminology....4
1 Scope....4
2 References....4
2.1 Normative references....4
2.2 Informative references....5
3 Definition of terms, symbols and abbreviations....5
3.1 Terms....5
3.2 Symbols....5
3.3 Abbreviations....5
3.4 Conventions....5
3.4.1 General....5
3.4.2 Requirements....5
3.4.3 UML diagrams....5
4 Business needs....6
4.1 Overview....6
4.2 Business needs for Service Registration....7
4.2.1 Register a service....7
4.2.2 Update a service registration....7
4.2.3 Query service registrations....7
4.2.4 De-register a service....7
4.3 Business needs for Service Discovery....8
4.3.1 Discover services....8
4.3.2 Manage subscriptions to service availability notifications....8
4.3.3 Notify service availability changes....8
4.4 Business needs for Authentication and authorization....9
4.4.1 Obtain authorization....9
4.4.2 Revoke authorization....9
4.5 Business needs for SME SMOS Bootstrap....9
4.5.1 Bootstrap....9
5 General aspects and principles....9
5.1 General aspects....9
5.2 Principles....10
6 Use cases....10
6.1 SME use case 1: Bootstrap....10
6.1.1 Overview....10
6.1.2 Background and goal of the use case....10
6.1.3 Entities/resources involved in the use case....10
6.1.4 Solutions....11
6.1.5 Required data....12
6.2 SME use case 2: Manage service registration....12
6.2.1 Overview....12
6.2.2 Background and goal of the use case....12
6.2.3 Entities/resources involved in the use case....12
6.2.4 Solutions....13
6.2.5 Required data....16
6.3 SME use case 3: Discover services....16
6.3.1 Overview....16
6.3.2 Background and goal of the use case....16
6.3.3 Entities/resources involved in the use case....17
6.3.4 Solutions....17

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6.3.5 Required data 18
6.4 SME use case 4: Subscribe service availability notifications 18
6.4.1 Overview 18
6.4.2 Background and goal of the use case 18
6.4.3 Entities/resources involved in the use case 18
6.4.4 Solutions 19
6.4.5 Required data 23
6.5 SME use case 5: Query registered services 23
6.5.1 Overview 23
6.5.2 Background and goal of the use case 23
6.5.3 Entities/resources involved in the use case 23
6.5.4 Solutions 24
6.5.5 Required data 25
7 Requirements 25
7.1 Security requirements 25
7.2 Functional requirements 25

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Foreword

This Technical Specification (TS) has been produced by WG2 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the general aspects, principles, business needs, use cases and requirements of the Service Management and Exposure (SME) SMO Service (SMOS).

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] O-RAN.WG1.TS.SMOARCH: "SMO Architecture" ("SMOARCH").

[2] OMG® Unified Modeling Language (OMG UML) version 2.5.1 ("OMG UML").

[3] Recommendation ITU-T M.3020: "Management interface specification methodology" ("ITU-T M.3020").

[4] O-RAN.WG11.TS.SRCS: "Security Requirements and Controls Specification" ("SRCS").

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2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [1] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [1] apply.

3.4 Conventions

3.4.1 General

For the purposes of the present document, the conventions in the following clauses apply.

3.4.2 Requirements

The requirements and solutions for use cases are based on the methodology specified in ITU-T M.3020 [3], clause A.1.2.

3.4.3 UML diagrams

The sequence diagrams for the messages that are exchanged between use case actors are based on the OMG UML [2], clause 17.4.4.1 and an example is illustrated in figure 3.4.3-1.

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Figure 3.4.3-1: Example of UML messages.

For each message in the present document, the following information is indicated:

- procedure name;
- type of message (e.g. request or response); and
- transferred information (if applicable).

4 Business needs

4.1 Overview

The business needs below are high level descriptions from the perspective of the consumer of SME services.

Variable Text	Description
Consumer	Who is requesting a capability of the SME SMOS
Feature Name	What capability is being consumed
Feature Behaviour	What extending behaviours does the feature need to provide, can be left blank
Expected Outcome	What is the expected outcome of the feature
Enabled Need	Why does the consumer need the feature

The variables can be inserted into an epic like story:

As the **{Consumer}** I need the SME Services to provide the capability to **{Feature Name}** [with the ability to **{Feature Behaviour}**] to **{Expected Outcome}** so that I can **{Enabled Need}**.

The SME SMOS allows the management and exposure of services, including SMOSs. It will fulfil the business needs of actors in the following roles:

- A Service Publisher, which wishes a service to be managed and exposed by the SME SMOS, that means to be discoverable and accessible by Service Consumers. Examples are the service producer itself, as well as an actor that determines which services are exposed
- A Service Consumer, which wishes to consume a service that is managed and exposed by the SME SMOS
- A Service Access Controller, which wishes to ensure that only authorized consumers can consume the service that is managed and exposed by the SME SMOS
- An SME SMOS Consumer, which wishes to consume the SME SMOS

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4.2 Business needs for Service Registration

4.2.1 Register a service

Variable	Value
Consumer	Service Publisher
Feature Name	Register service
Feature Behaviour	
Expected Outcome	A service has been registered, such that the service can be discovered and authorized.
Enabled Need	The Service Publisher needs to make information about the exposed services available such that they can be discovered by authorized Service Consumers.

4.2.2 Update a service registration

Variable	Value
Consumer	Service Publisher
Feature Name	Update service registration
Feature Behaviour	
Expected Outcome	The information about a registered service has been updated.
Enabled Need	The Service Publisher needs to be able to update the information about its registered services in case of changes.

4.2.3 Query service registrations

Variable	Value
Consumer	Service Publisher
Feature Name	Query service registrations
Feature Behaviour	
Expected Outcome	The information about registered services has been made available.
Enabled Need	The Service Publisher needs to understand which SMOSs have already been registered in order to recover from failures.

4.2.4 De-register a service

Variable	Value
Consumer	Service Publisher
Feature Name	De-register service
Feature Behaviour	
Expected Outcome	The registration of a service and its producer has been removed, such that the service can no longer be discovered and authorized.

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Enabled Need	The Service Publisher needs to remove information about services no longer available such that they can't be discovered by consumers anymore.

4.3 Business needs for Service Discovery

4.3.1 Discover services

Variable	Value
Consumer	Service Consumer
Feature Name	Provide a list of services that can be consumed by an authorized Service Consumer.
Feature Behaviour	
Expected Outcome	A list of services that can be consumed by an authorized Service Consumer is available.
Enabled Need	A Service Consumer needs to obtain information about the services available for consumption.

4.3.2 Manage subscriptions to service availability notifications

Variable	Value
Consumer	Service Consumer
Feature Name	Manage subscriptions to service availability notifications
Feature Behaviour	
Expected Outcome	A subscription to service availability notifications has been created, read, updated or deleted.
Enabled Need	A Service Consumer needs to indicate that it wishes to receive or stop receiving notifications related to changes in SMOS availability.

4.3.3 Notify service availability changes

Variable	Value
Consumer	Service Consumer
Feature Name	Notify service availability changes
Feature Behaviour	
Expected Outcome	Notifications about availability of services have been sent to subscribed entities.
Enabled Need	A Service Consumer needs to have up-to-date information about service availability.

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4.4 Business needs for Authentication and authorization

4.4.1 Obtain authorization

Variable	Value
Consumer	Service Consumer
Feature Name	Obtain authorization
Feature Behaviour	
Expected Outcome	The Service Consumer is authorized to consume a service.
Enabled Need	Ensure that only authorized Service Consumers can consume services.

Authorization requires the prior authentication of Service Consumers and Service Producers.

4.4.2 Revoke authorization

Variable	Value
Consumer	Service Access Controller
Feature Name	Revoke authorization
Feature Behaviour	
Expected Outcome	A particular Service Consumer is not allowed to consume a service any longer.
Enabled Need	Exercise control over the allowed consumers of a service at any time.

4.5 Business needs for SME SMOS Bootstrap

4.5.1 Bootstrap

Variable	Value
Consumer	SME SMOS Consumer
Feature Name	Bootstrap service discovery
Feature Behaviour	
Expected Outcome	An SME SMOS Consumer knows the SME SMOS entry point information needed to start discovering services.
Enabled Need	Allow SME SMOS Consumers to get started with the discovery of services from a zero-knowledge state.

5 General aspects and principles

5.1 General aspects

The SME SMOS enables the exposure of services (including SMOSs) by supporting capabilities to:

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- Register services,
- Discover services,
- Authenticate and authorize Service Consumers,
- Bootstrap (optional).

The bootstrap functionality allows an SME SMOS consumer to discover the endpoints of the SME SMOS, if such endpoints have not been made known to the SME SMOS consumers by other means.

The SME SMOS is able to handle services that are

- produced by rApps in the NonRT RIC,
- consumed by authorized rApps in the NonRT RIC,
- produced by SMOFs,
- consumed by authorized SMOFs.

Additionally, the SME SMOS may allow authorized SMOFs and authorized rApps to consume services that are produced by Service Producers outside the SMO framework and may allow authorized Service Consumers outside the SMO framework to consume services that are produced by Service Producers inside the SMO framework.

The SME SMOS also handles the authorization of Service Consumers.

5.2 Principles

Only authorized SME SMOS consumers are allowed to register services.

Only authorized SME SMOS consumers are allowed to discover services.

Only authorized Service Consumers are allowed to consume the services.

Once a service is registered, it can be made available for discovery and invocation.

6 Use cases

6.1 SME use case 1: Bootstrap

6.1.1 Overview

This use case allows an SME SMOS consumer to retrieve bootstrap information that enables it to optionally discover the SME SMOS endpoints for registering and discovering services.

6.1.2 Background and goal of the use case

The SME SMOS consumer can discover the SME SMOS endpoints for registering and discovering services by retrieving the bootstrap information.

6.1.3 Entities/resources involved in the use case

1) SME SMOS producer

a) supports functionality that allows the SME SMOS consumer to retrieve the bootstrap information.

2) SME SMOS consumer

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a) retrieves the bootstrap information.

6.1.4 Solutions

6.1.4.1 Retrieve bootstrap information

Table 6.1.4.1-1: Retrieve bootstrap information use case

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	Find the SME SMOS endpoints.	
Actors and Roles	SME SMOS consumer that obtains the bootstrap information. SME SMOS producer that provides the bootstrap information.	
Assumptions	n/a	
Preconditions	SMOS producer and SMOS consumer are instantiated.	
Begins when	The SMOS consumer has determined the need to register or discover services.	
Step 1	The SMOS consumer will initiate a bootstrap request to find the SME SMOS endpoints.	
Step 2	The SME SMOS producer responds with the bootstrap information that contains details of the SME SMOS service endpoints.	
Ends when	The SME SMOS consumer has retrieved the bootstrap information that contains details of the SME SMOS service endpoints.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer has retrieved the bootstrap information that contains details of the SME SMOS service endpoints.	
Traceability	REQ-SME-SMOS-Boot-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box  #cadetBlue
     participant  "SME SMOS producer"  as prod
endbox
cons -> prod : Bootstrap request
activate prod
prod --> cons : Bootstrap response
deactivate prod
@enduml
```

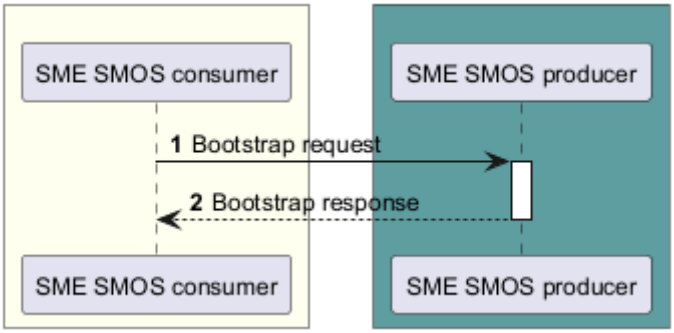


Figure 6.1.4.1-1: Retrieve bootstrap information use case flow diagram

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6.1.5 Required data

The SME SMOS consumer may obtain information about the SME SMOS endpoints from the SME SMOS producer.

6.2 SME use case 2: Manage service registration

6.2.1 Overview

This use case allows an SME SMOS consumer to register services, update the service information of each of its registered services and deregister each of its registered services.

Often, the functional entity which registers the produced service is the service producer itself, however, it can also be a functional entity that has the purpose of managing the service registration on behalf of the service producers.

6.2.2 Background and goal of the use case

A SME SMOS consumer can register services. Other SME SMOS consumers can discover the registered services.

After an SME SMOS consumer has successfully registered services, it can update and/or deregister each of the registered services.

6.2.3 Entities/resources involved in the use case

1) SME SMOS producer that produces the SME SMOS to register services:
    a. supports functionality to register a service produced by a service producer,
    b. supports authorization of an SME SMOS producer to determine whether it can register a service, update a registered service, and deregister a registered service,
    c. supports validation of the service to be registered that includes determining whether there are conflicts with services registered earlier,
    d. responds to the requests to register a service, update a service registration and deregister a service, indicating the success or failure result
2) SME SMOS consumer that registers a service
    a. supports to request the registration of a service, the update of a registered service, and the deregistration of a registered service.

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6.2.4 Solutions

6.2.4.1 Register service

Table 6.2.4.1-1: Register service use case

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer successfully registers a service.	
Actors and Roles	SME SMOSS consumer that requests registration of a service. SME SMOS producer that handles the service registration.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is deployed and authorized to register a service and optionally has the bootstrap info.	
Begins when	The SME SMOS consumer determines the need to register a service.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to register a service by providing its identifier and the service profile. That identifier identifies the SME SMOS consumer as an entity that is entitled to register services.	
Step 2 (M)	The SME SMOS producer checks whether the SME SMOS consumer is authorized to register a service.	
Step 3 (M)	The SME SMOS producer validates the information about the service to be registered, that includes determining whether there are conflicts with services registered earlier.	
Step 4 (M)	The SME SMOS producer registers the service and makes the service endpoints available for discovery.	
Step 5 (M)	The SME SMOS producer responds to the SME SMOS consumer with a success result along with a service identifier.	
Ends when	The SME SMOS consumer was able to register a service.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer will be able to update and de-register the registered service.	
Traceability	REQ- SME-SMOS-ServiceReg-FUN1, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box #cadetBlue
     participant  "SME SMOS producer"  as prod
endbox
cons->prod: Register service request (consumer identifier, service profile)
activate prod
  prod --> prod: AuthZ
  note right
  check authorization to
  register a service
  end note
  prod --> prod: Validate
  prod --> prod: register service
  prod --> cons: Register service response (service identifier)
deactivate prod
@enduml
```

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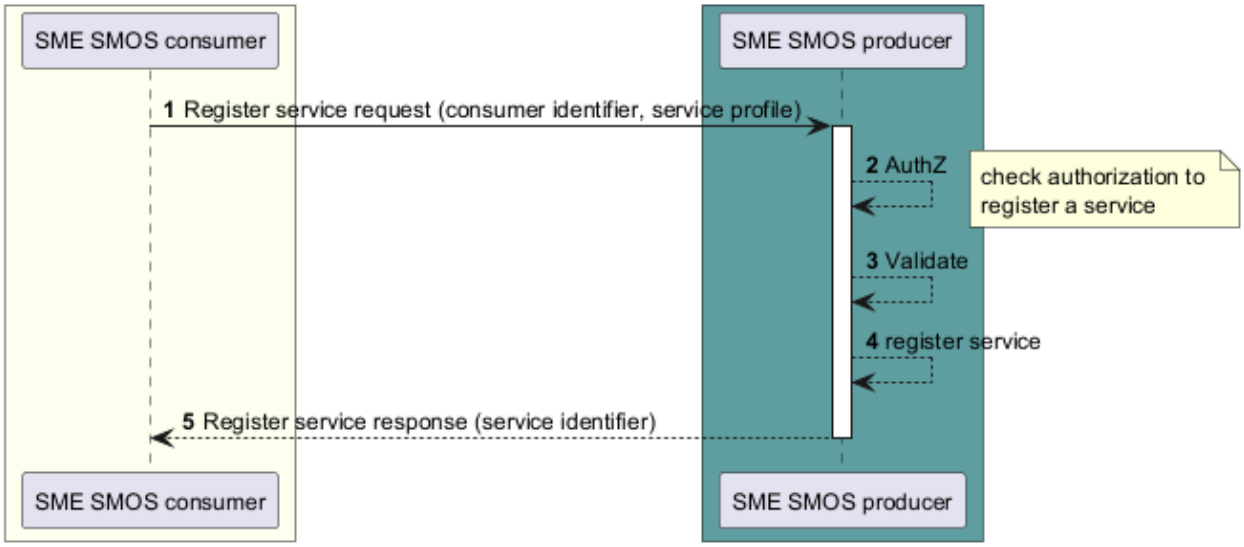


Figure 6.2.4.1-1: Register service use case flow diagram.

6.2.4.2 Update service registration

Table 6.2.4.2-1: Update service registration use case.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer successfully updates a service registration.	
Actors and Roles	SME SMOS consumer that requests updating the registered information of a service. SME SMOS producer that handles the update to the service registration.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to update a service registration. The SME SMOS consumer has registered a service.	
Begins when	The SME SMOS consumer determines the need to update the registered information of a service.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to update the registered information of a service by providing its service consumer identifier, service identifier and delta service profile.	
Step 2 (M)	The SME SMOS producer checks whether the SME SMOS consumer is authorized to update a registered service.	
Step 3 (M)	The SME SMOS producer updates the service information and makes the updated service endpoint available for discovery.	
Step 4 (M)	The SME SMOS producer responds with a successful result.	
Ends when	The SME SMOS consumer was able to update service registration.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer will be able to update and de-register the registered service.	
Traceability	REQ- SME-SMOS-ServiceReg-FUN2, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
  box #ivory
    participant " SME SMOS consumer" as cons
  endbox
  box  #cadetBlue
```

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```
    participant  "SME SMOS producer"  as prod
endbox
cons -> prod: Update service registration request (consumer identifier,service identifier,
service profile update)
activate prod
  prod --> prod: AuthZ
note right
check authorization to
update a registered service
end note
  prod --> prod: service update
  prod --> cons: Update service registration response
deactivate prod
@enduml
```

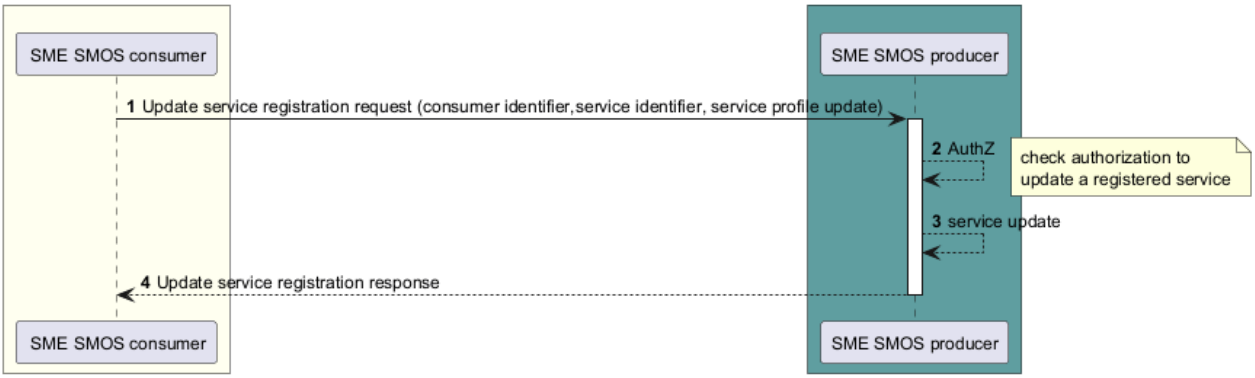


Figure 6.2.4.2-1: Update service registration use case flow diagram.

6.2.4.3 Deregister Service

Table 6.2.4.3-1: Deregister service use case.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer successfully deregisters a service.	
Actors and Roles	SME SMOSS consumer that requests service deregistration. SME SMOS producer that handles the service deregistration.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to deregister a service. The SME SMOS consumer has previously registered the service that is requested to be deregistered.	
Begins when	The SME SMOS consumer determines the need to deregister a registered service.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to deregister a service by providing the consumer identifier and service identifier.	
Step 2 (M)	The SME SMOS producer checks if the SME SMOS consumer is authorized to deregister a service.	
Step 3 (M)	The SME SMOS producer removes the registered service and removes the service endpoint from the set of endpoints that can be discovered.	
Step 4 (M)	The SME SMOS producer responds with a successful result.	
Ends when	The SME SMOS consumer was able to deregister a service.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer has deregistered a registered service; service endpoint will be no longer available for service discovery.	
Traceability	REQ- SME-SMOS-ServiceReg-FUN3, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
```

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```
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
  box #cadetBlue
participant  "SME SMOS producer" as SME
endbox
cons -> SME: Deregister service request (consumer identifier,service identifier)
activate SME
SME --> SME: AuthZ
note right
check authorization to
deregister a registered service
end note
  SME --> SME: Remove service profile
  SME --> cons: Deregister service response
deactivate SME
@enduml
```

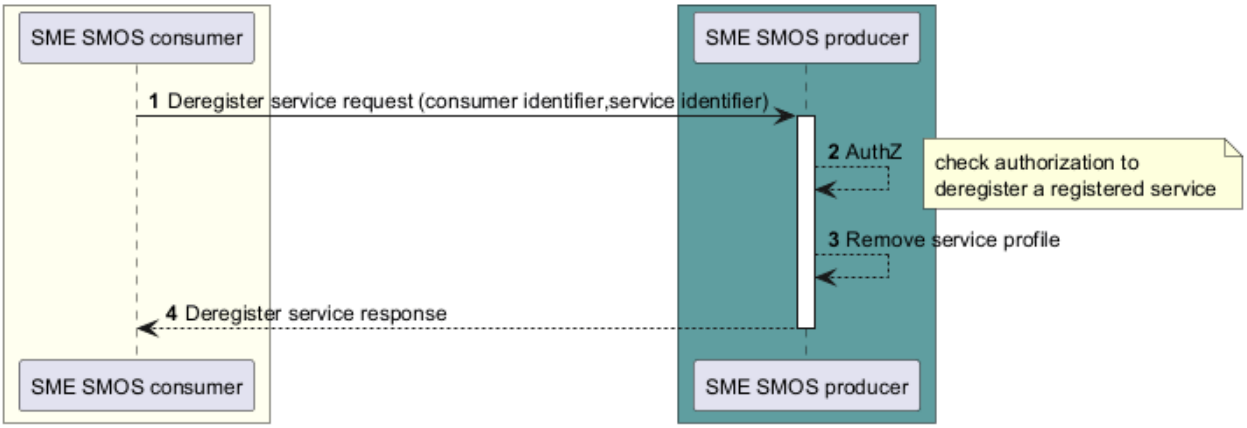


Figure 6.2.4.3-1: Deregister service use case flow diagram.

6.2.5 Required data

The service profile information includes e.g., the name of the service, service type, communication mechanism, interface endpoint details (such as IP address, port number, URI, protocol), version number, and data format.

For registering a service, the SME SMOS consumer needs to provide the consumer identifier and the service profile. On successful registration, the SME SMOS producer assigns a service identifier for the newly registered service and returns it to the SME SMOS consumer.

For updating a registered service, the SME SMOS consumer needs to provide the consumer identifier, the service identifier and the delta information that needs to be changed in the service profile.

For de-registering a registered service, the SME SMOS consumer needs to provide the consumer identifier and the service identifier.

6.3 SME use case 3: Discover services

6.3.1 Overview

This use case allows an SME SMOS consumer to discover the available services.

6.3.2 Background and goal of the use case

An SME SMOS consumer will be able to discover the available services.

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6.3.3 Entities/resources involved in the use case

1) SME SMOS producer
    a. supports functionality to discover the available services,
    b. supports authorization of an SME SMOS consumer to determine which services it can discover,
    c. retrieves the stored service information and performs filtering of available services based on selection criteria that may be provided by the SME SMOS consumer,
    d. responds to the request to discover the available services, indicating the success or failure result.
2) SME SMOS consumer
    a. supports requesting the discovery of the available services, optionally providing selection criteria.

6.3.4 Solutions

6.3.4.1 Discover the services

Table 6.3.4.1-1: Discover services use case.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer discovers the available services.	
Actors and Roles	SME SMOS consumer that requests to discover the available services. SME SMOS producer that handles the discovery of available services.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to discover services.	
Begins when	The SME SMOS consumer determines the need to discover the available services.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to discover available services based on the consumer identifier and optional selection criteria.	
Step 2 (M)	The SME SMOS producer checks which services the SME SMOS consumer is authorized to discover.	
Step 3 (M)	The SME SMOS producer responds with the set of available services and their service identifiers. The list contains only those available services that match the selection criteria if those were provided by the consumer, or all available services otherwise.	
Ends when	The SME SMOS consumer was able to discover the available services.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer can request access to the available services.	
Traceability	REQ-SME-SMOS-ServiceDiscovery-FUN1, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
 'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box #cadetBlue
     participant  "SME SMOS producer" as SME
endbox
cons -> SME: Discover services request(consumer identifier)
activate SME
SME --> SME: AuthZ
```

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```
note right
check authorization to
discover available services
end note
SME --> cons: Discover services response(list of services)
deactivate SME
@enduml
```

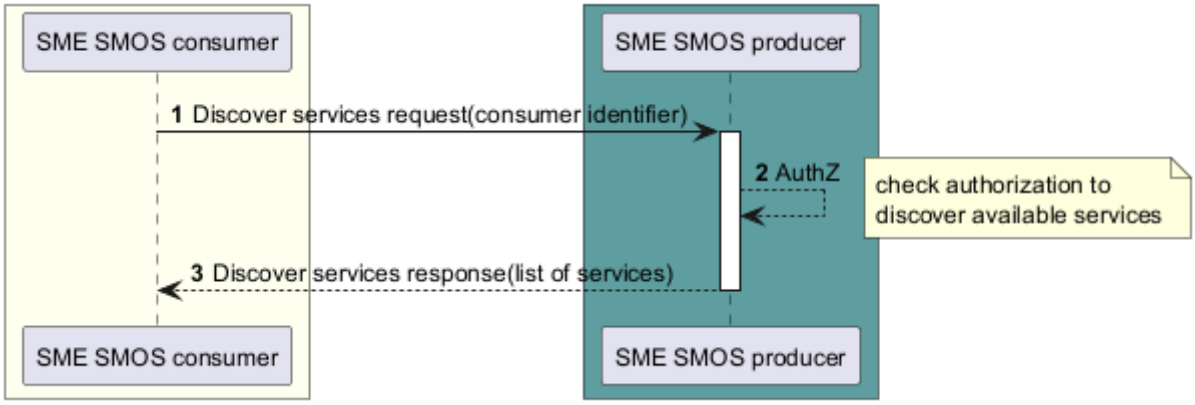


Figure 6.3.4.1-1: Discover services use case flow diagram.

6.3.5 Required data

For discovering the available services, the SME SMOS consumer needs to provide the consumer identifier and optional selection criteria such as e.g., name of the service, service type, and capabilities.

The SME SMOS producer responds with information about a set of services which includes e.g., endpoint information and service identifiers.

NOTE: Whether an SME SMOS consumer is allowed to discover services for which it is currently not authorized but might be able to obtain authorization later is not defined in the present document.

6.4 SME use case 4: Subscribe service availability notifications

6.4.1 Overview

This use case allows an SME SMOS consumer to subscribe to notifications and unsubscribe from notifications regarding changes in service availability, to receive notifications about subscribed events and to update its subscriptions.

6.4.2 Background and goal of the use case

The SME SMOS consumer can subscribe to notifications, unsubscribe from notifications and receive notifications regarding changes in the availability of services. Further, it can update its subscriptions.

6.4.3 Entities/resources involved in the use case

1) SME SMOS producer

   a) supports functionality allowing an SMS SMOS consumer to subscribe to and unsubscribe from service availability notifications as well as to update its subscriptions,

   b) supports notifying the SME SMOS consumer regarding changes in the availability of services,

   c) responds to the requests to subscribe service availability, to update a service availability subscription and to unsubscribe service availability, indicating the success or failure result.

2) SME SMOS consumer

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a) supports functionality to initiate requests to subscribe to and unsubscribe from service availability notifications as well as to update its subscriptions,

b) supports functionality to accept notifications of the changes in service availability from the SME SMOS producer.

6.4.4 Solutions

6.4.4.1 Subscribe to service availability notifications

Table 6.4.4.1-1: Subscribe to service availability notifications.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer subscribes to notifications about changes of the availability of services.	
Actors and Roles	SME SMOS consumer that requests subscription to service availability notifications. SME SMOS producer that handles the subscription requests.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to subscribe to notifications about changes in service availability.	
Begins when	The SME SMOS consumer determines to subscribe to notifications regarding changes in the availability of services.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to subscribe to notifications regarding changes in the available services with the consumer identifier and optional service identifiers.	
Step 2 (M)	The SME SMOS producer checks whether the SME SMOS consumer is authorized to send a subscription request.	
Step 3 (M)	The SME SMOS producer responds with the subscription identifier.	
Ends when	The SME SMOS consumer was able to subscribe to notifications.	
Exceptions	n/a	
Post Conditions	The SME SMOS consumer can receive notifications when there are any changes in the service availability. The SME SMOS consumer can unsubscribe from service availability notifications.	
Traceability	REQ- SME-SMOS-ServiceSub-FUN1, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
  box #cadetBlue
participant "SME SMOS producer"  as SME
endbox
autonumber
cons -> SME: Subscribe service availability request(consumer identifier)
activate SME
 SME --> SME: AuthZ
note right
check authorization to
send a subscription request
end note
   SME --> cons: Subscribe service availability response (subscription identifier)
deactivate SME
@enduml
```

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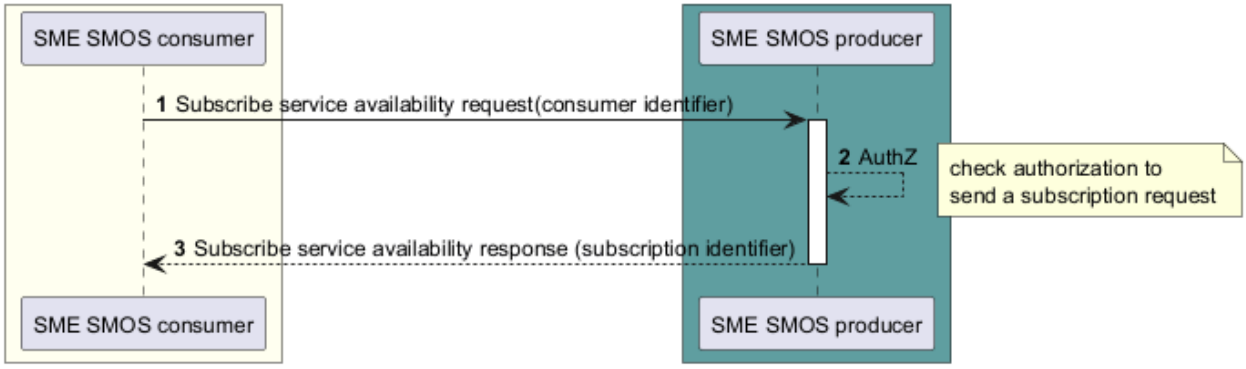


Figure 6.4.4.1-1: Subscribe to service availability notifications use case flow diagram.

6.4.4.2 Notify service availability changes

Table 6.4.4.2-1: Notify service availability changes use case.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer receives notifications about changes in the available services.	
Actors and Roles	SME SMOS consumer that receives notifications on service availability changes. SME SMOS producer that sends notifications on service availability changes.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer has subscribed to notifications on service availability changes.	
Begins when	The SME SMOS producer determines to send notifications regarding changes in the available services to the subscribed consumer.	
Step 1 (M)	The SME SMOS producer detects changes in service availability (e.g. due to service registration, service update or service deregistration).	
Step 2 (M)	The SME SMOS producer sends notifications with information about the service changes to the subscribed SME SMOS consumer with subscription identifier.	
Step 3 (M)	The SME SMOS consumer processes the changes.	
Ends when	The SME SMOS consumer was able to process the changes in the available services.	
Exceptions	n/a	
Post Conditions	n/a	
Traceability	REQ-SME-SMOS-ServiceSub-FUN2.	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant " SME SMOS consumer" as cons
endbox
box #cadetBlue
     participant  "SME SMOS producer"  as SME
endbox
autonumber
SME --> SME: Trigger changes in the services
SME ->> cons: Notify changes \n(subscription identifier, service identifiers)
cons --> cons: Process the changes
@enduml
```

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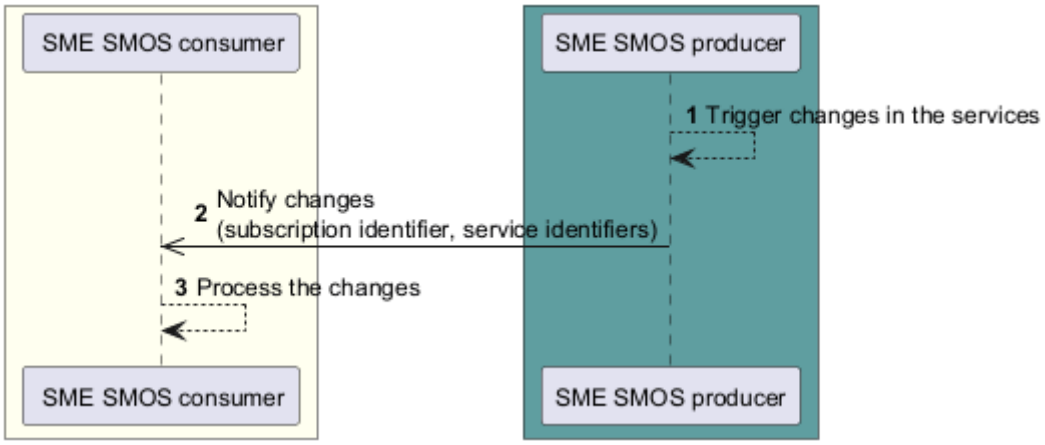


Figure 6.4.4.2-1: Notify service availability changes use case flow diagram.

6.4.4.3 Unsubscribe from service availability notifications

Table 6.4.4.3-1: Unsubscribe from service availability notifications.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer unsubscribes from service availability notifications.	
Actors and Roles	SME SMOS consumer that requests unsubscribe from service availability notifications. SME SMOS producer that processes the unsubscribe service availability request.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to unsubscribe from service availability notifications. The SME SMOS consumer has subscribed to service availability notifications.	
Begins when	The SME SMOS consumer determines to unsubscribe from service availability notifications.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to unsubscribe from service availability notifications with consumer identifier and subscription identifier.	
Step 2 (M)	The SME SMOS producer checks whether the SME SMOS consumer is authorized to unsubscribe.	
Step 3 (M)	The SME SMOS producer sends a response.	
Ends when	The SME SMOS consumer was able to unsubscribe from service availability notifications.	
Exceptions	n/a	
Post Conditions	The SME SMOS producer is not subscribed to service availability notifications.	
Traceability	REQ-SME-SMOS-ServiceSub-FUN3.	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box #cadetBlue
     participant  "SME SMOS producer"  as SME
endbox
autonumber
cons -> SME : Unsubscribe from service availability notifications request(subscription
identifier)
```

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```
activate SME
   SME --> SME: AuthZ
note right
check authorization to
unsubscribe
end note
   SME --> cons: Unsubscribe from service availability notifications response
deactivate SME
@enduml
```

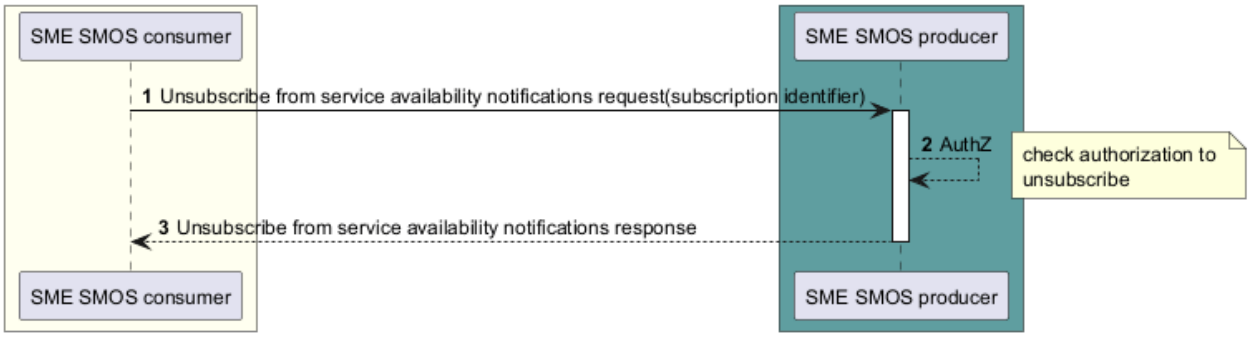


Figure 6.4.4.3-1: Unsubscribe service availability use case flow diagram.

6.4.4.4 Update subscriptions to service availability notifications

Table 6.4.4.4-1: Update subscriptions to service availability notifications

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer updates its subscriptions to notifications about changes in the availability of services.	
Actors and Roles	SME SMOS consumer that requests updating one of its subscriptions to service availability notifications. SME SMOS producer entity that handles the update subscription requests.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to update its subscriptions to notifications about changes in service availability.	
Begins when	The SME SMOS consumer determines to update one of its subscriptions to notifications regarding changes in the availability of services.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to update one of its subscriptions to notifications regarding changes in the available services with consumer identifier and subscription identifier.	
Step 2 (M)	The SME SMOS producer checks whether the SME SMOS consumer is authorized to update the subscription.	
Step 3 (M)	The SME SMOS producer sends a response.	
Ends when	The SME SMOS consumer was able to update one of its subscriptions to notifications.	
Exceptions	n/a	
Post Conditions	None	
Traceability	REQ-SME-SMOS-ServiceSub-FUN, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box #cadetBlue
```

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```
     participant  "SME SMOS producer"  as SME
endbox
autonumber
cons -> SME: Update service availability subscription request(subscription identifier,
subscription update information)
activate SME
SME --> SME: AuthZ
note right
check authorization to
update the subscription
end note
SME --> cons: Update service availability subscription response
deactivate SME
@enduml
```

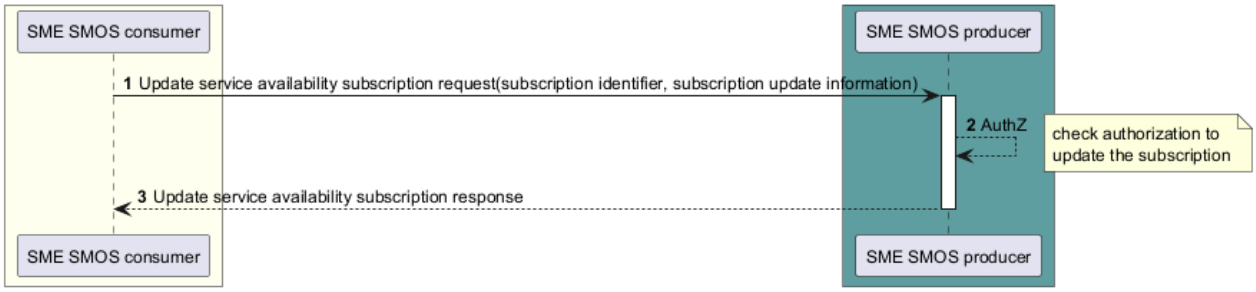


Figure 6.4.4.1-1: Update subscriptions to service availability notifications use case flow diagram.

6.4.5 Required data

To receive notifications regarding changes in service availability, the SME SMOS consumer subscribes to notifications by providing the consumer identifier and optional service identifiers. When the SME SMOS producer has successfully processed the request, it responds by providing a subscription identifier.

The SME SMOS producer sends notifications to the subscribed SME SMOS consumer when there are changes in the availability of services by providing subscription identifier, service identifiers and information about the changes.

For updating a subscription, a subscribed SME SMOS consumer needs to provide the subscription identifier and subscription update information.

For unsubscribing from notifications, a subscribed SME SMOS consumer needs to provide subscription identifier.

6.5 SME use case 5: Query registered services

6.5.1 Overview

This use case allows an SME SMOS consumer to retrieve the service information of services it has registered.

6.5.2 Background and goal of the use case

An SME SMOS consumer can retrieve the service information of its registered services from the SME SMOS producer.

6.5.3 Entities/resources involved in the use case

1) SME SMOS producer

   a. supports functionality to query the information of registered services,

   b. supports authorization of an SME SMOS consumer to determine whether the consumer can query the information of registered services,

   c. retrieves the stored service information and performs filtering of available services based on the selection criteria that may be provided by SME SMOS consumer,

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d. responds to the request to query registered services, indicating the success or failure result.

2) SME SMOS consumer

a. supports to initiate the procedure to query the service information of its registered services.

6.5.4 Solutions

6.5.4.1 Query registered services

Table 6.5.4.1-1: Query registered services use case.

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	The SME SMOS consumer is aware of the service information of its registered services.	
Actors and Roles	SME SMOS consumer that queries registered services. SME SMOS producer that handles the queries for registered services.	
Assumptions	n/a	
Preconditions	The SME SMOS consumer is authorized to query its registered services.	
Begins when	The SME SMOS consumer determines the need to query registered services.	
Step 1 (M)	The SME SMOS consumer requests the SME SMOS producer to query registered services based on the consumer identifier and optional selection criteria.	
Step 2 (M)	The SME SMOS producer checks if the SME SMOS consumer is authorized to query registered services.	
Step 3 (M)	The SME SMOS producer responds with service profiles that match the query.	
Ends when	The SME SMOS consumer was able to query registered services.	
Exceptions	n/a	
Post Conditions	n/a	
Traceability	REQ-SME-SMOS-ServiceRegQuery-FUN1, REQ-SME-SMOS-Authz-FUN1	

```
@startuml
 'https://plantuml.com/sequence-diagram
 !pragma teoz true
 skinparam ParticipantPadding 5
 skinparam BoxPadding 10
 skinparam defaultFontSize 12
 skinparam lifelineStrategy solid
 autonumber
box #ivory
     participant "SME SMOS consumer" as cons
endbox
box #cadetBlue
     participant  "SME SMOS producer" as SME
endbox
cons -> SME: Query registered services request (consumer identifier)
activate SME
SME --> SME: AuthZ
note right
check authorization to
query registered services
end note
SME --> cons: Query registered services response (service profiles)
deactivate SME
@enduml
```

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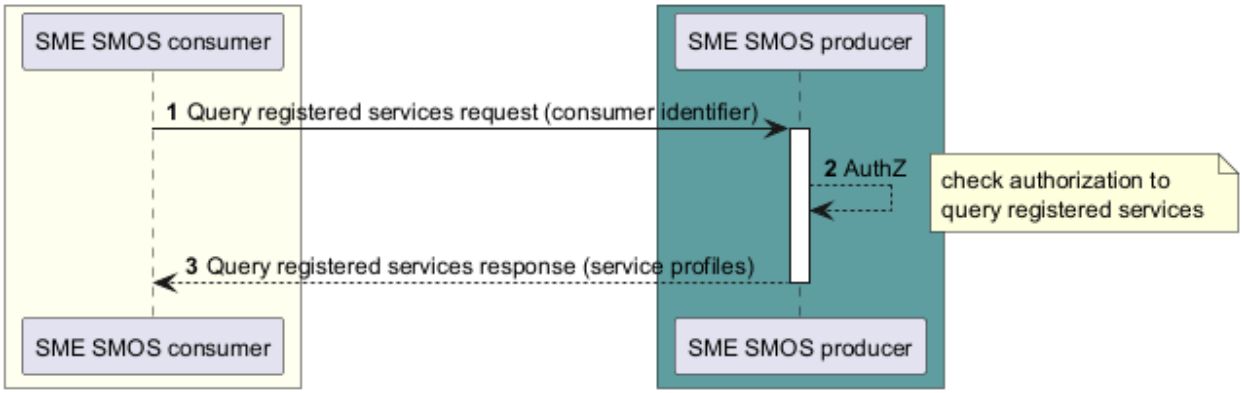


Figure 6.5.4.1-1: Query registered services use case flow diagram.

6.5.5 Required data

For querying registered services, the SME SMOS consumer needs to provide the consumer identifier and optional selection criteria e.g., service name, service type and capabilities.

The SME SMOS producer responds with service profiles that match the selection criteria.

7 Requirements

7.1 Security requirements

The security requirements defined in clause 5.1.1.1.6 of SRCS [4] apply.

7.2 Functional requirements

This clause specifies the functional requirements for the SME SMOS in table 7.2-1.

Table 7.2-1: SME SMOS requirements

REQ	Description	Note
REQ-SME-Authz-FUN1	The SME SMOS shall support the authorization of a service consumer.	
REQ-SME-ServiceReg-FUN1	The SME SMOS shall support the registration of a service.	
REQ-SME-ServiceReg-FUN2	The SME SMOS shall support the update of a service registration.	
REQ-SME-ServiceReg-FUN3	The SME SMOS shall support the deregistration of a registered service.	
REQ-SME-ServiceDiscovery-FUN1	The SME SMOS shall support the discovery of services.	
REQ-SME-ServiceSub-FUN1	The SME SMOS shall support the subscription to notifications of service availability changes.	
REQ-SME-ServiceSub-FUN2	The SME SMOS shall support sending notifications for service availability changes.	
REQ-SME-ServiceSub-FUN3	The SME SMOS shall support unsubscription from notifications of service availability changes.	
REQ-SME-ServiceRegQuery-FUN1	The SME SMOS shall support querying service registrations.	
REQ-SME-Boot-FUN1	The SME SMOS should support functionality to obtain bootstrap information.	

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Annex (informative): Change history

Date	Revision	Description
2025-10-20	00.00.01	Initial version.
2025-11-27	00.00.02	Incorporated: • NOK.AO-2025.11.12-WG2-CR-203-SME_SMOS_Apply_WG1_Dallas_approved_template-v03
2026-02-13	00.00.03	Incorporated: • NOK-2026.01.23-WG2-CR-0206-SME_4 Business needs_v03 • NOK-2026.01.23-WG2-CR-0207-SME_5.1 General aspects_v03 • NOK-2026.01.23-WG2-CR-0208-SME_5.2 Principles_v02 • NOK-2026.01.23-WG2-CR-0210-SME_SMOS_6 Use cases_v02 • NOK-2026.01.23-WG2-CR-0211-SME_7 Requirements_v03
2026-03-05	00.00.04	Incorporated: • NOK-2026.02.03-WG2-CR-0213-SME_Expand abbreviations in Scope clause_v01 • NOK-2026.02.24-WG2-CR-0214-SME_Security Requirements reference_v02
2026-03-12	01.00	Addressed comments in editorial review. This version to be published in March 2026 train

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